

2nd Year 4th Semester									
Sl. No.	Broad Category	Category	Course Code	Course Title	Hours per week				Credits
					L	T	P	Total	
A. THEORY									
1	ENGG	Major	AM401	Operating Systems	3	0	0	3	3
2	ENGG	Major	AM402	Formal Language and Automata Theory	3	0	0	3	3
3	ENGG	Major	AM403	Object Oriented Programming using Java	3	1	0	3	4
4	SCI	Minor	M(AM)401	Probability and Statistics	3	0	0	3	3
5	HUM	Ability Enhancement Course	HU(AM)401	Principles of Management	2	0	0	2	2
B. PRACTICAL									
1	ENGG	Major	AM491	Operating Systems Lab	0	0	3	3	1.5
2	ENGG	Minor	M(AM)492	Numerical Methods Lab	0	0	3	3	1.5
3	ENGG	Major	AM493	Object Oriented Programming Lab using Java	0	0	3	3	1.5
4	ENGG	Major	AM494	Programming Using Python	0	0	3	3	1.5
5	HUM	Ability Enhancement Course	HU(AM)491	Soft Skill & Aptitude	2	0	0	2	1
Total of Theory and Practical								27	22

Paper Name: Operating System

Paper Code: AM401

Contact Hours/Week: 3

Credit: 3

Total Contact Hours: 36L

Prerequisites:

1. Computer organization
2. Computer Architecture
3. Data Structures
4. Algorithms & Programming Concept

Course Outcomes (COs):

After attending the course students should be able to

CO1	Understand the fundamental concepts of Operating System, Protection & Security and differentiate different types of Operating System.
CO2	Understand and implement process & thread; understand, apply, compare different process synchronization algorithm and inter process communication to solve engineering problems
CO3	Understand/explain/analyze different synchronization techniques, critical section problems and deadlock and apply them to solve engineering problems.
CO4	Understand/explain different memory management techniques including virtual memory management; also able to apply, compare, and implement different page replacement algorithms to solve engineering problems.
CO5	Understand/explain different I/O mechanisms, File structures and disk management techniques and solving engineering problems applying different disk scheduling algorithms.

Course Content:

Module – 1: [3L]

Functionalities of Operating System, Evolution of Operating System.

Types of Operating System: batch, multi-programmed, time-sharing, real-time, distributed, parallel, Structural overview, Protection & Security. [3L]

Module – 2: [11L]

Processes: Concept of processes, process states, PCB, process scheduling, co-operating processes, independent process, suspended process, Interaction between processes and OS, Inter-process communication: Message passing. [3L]

Threads: overview, benefits of threads, user and kernel level threads, Thread models. [2L]

CPU scheduling: Scheduling criteria, preemptive & non-preemptive scheduling, scheduling algorithms (FCFS, SJF, SRTF, RR, priority, multilevel queue, multilevel feedback queue scheduling). [6L]

Module – 3: [11L]

Process Synchronization: background, critical section problem, synchronization hardware, classical problems of synchronization (producer-consumer, readers-writer, dining philosophers, etc), semaphores, monitors.[6L]

Deadlocks: deadlock characterization, methods for handling deadlocks, deadlock prevention,

deadlock avoidance, deadlock detection, recovery from deadlock.[5L]

Module 4: [6L]

Background, logical vs. physical address space, swapping, contiguous memory allocation, paging, Segmentation, TLB. [3L]

Virtual Memory: background, demand paging, page replacement algorithms (FCFS, LRU, Optimal), thrashing, Working set model. [3L]

Module 5: [5L]

Disk structure, disk scheduling (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK etc), disk reliability, disk formatting, boot block, bad blocks. [2L]

File: File concept, access methods, directory structure, file system structure, UNIX file structure, allocation methods (contiguous, linked, indexed), free-space management (bit vector). [2L]

I/O: I/O hardware, polling, interrupts, DMA, caching, buffering, blocking-non blocking I/O. [1L]

Text Book:

1. Abraham Silberschatz, Peter B. Galvin, Greg Gagne, Operating System Concepts.
2. Operating Systems & Systems Programming by P Balakrishna Prasad

Reference Book:

1. Dietel H. N., —An Introduction to Operating Systems, Addison Wesley.
2. Andrew Tanenbaum, Modern Operating Systems, Prentice Hall.
3. William Stallings, Operating Systems, Prentice Hall.

CO-PO Mapping

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								3
CO2	3	3	3	3								3
CO3	3	3	3	3								3
CO4	3	3	3	3								3
CO5	3	3	3	3								3

CO-PSO Mapping

COs	PSO1	PSO2	PSO3
CO1	3	3	3
CO2	3	3	3
CO3	3	3	3
CO4	3	3	3
CO5	3	3	3

Formal Language and Automata Theory**Course Code: AM402****Contacts: 3:0:0****Total Contact Hours: 36****Credits: 3****Prerequisites:**

1. Digital Logic
2. Computer organization
3. Computer Fundamentals

Course Outcomes (COs):

After attending the course students should be able to

CO1	Understand the fundamental concepts of Finite State Automata to Explain or Illustrate and Identify problems where students can Apply the concept appropriately to Solve them.
CO2	Understand the fundamental concepts of Regular Expressions and its relation with DFA so that they can Develop regular expression for a specified language and Validate it.
CO3	Understand the fundamental concepts of Context Free Grammar so that they can Design grammar for a specified language and Validate it.
CO4	Explain or Illustrate the fundamental operating principles of Push Down Automata and Use it appropriately to Solve problems.
CO5	Understand the operating principles of Turing Machine and Design Turing Machines to Propose solutions to the related problems appropriately and validate the effectiveness as well as limitations of computations making the students aware of its utilitarian importance for further explorations leading towards lifelong learning.

Course Contents:**Module-1: [9L]**

Fundamentals: Basic definition of sequential circuit, block diagram, mathematical representation, concept of transition table and transition diagram, [1L]

Introduction to Finite State Model (FSM), Design of sequence detector, Finite State Machine, Finite Automata, Deterministic Finite Automata (DFA) and Non-deterministic Finite Automata (NFA), Transition diagrams, Transition tables and Language recognizers.[3L]

NFA with empty transitions, Equivalence between NFA with and without empty transitions. NFA to DFA conversion. [2L]

Minimization of FSM: Minimization Algorithm for DFA, Introduction to Myhill-Nerode Theorem [2L] Limitations of FSM, Application of Finite Automata[1L]

Module-2: [7L]

Finite Automata with output – Moore & Mealy machine. Representation of Moore & Mealy Machine, Processing of the String through Moore & Mealy Machine, Equivalence of Moore & Mealy Machine – Inter- conversion. [2L]

Equivalent states and Distinguishable States, Equivalence and k-equivalence, Minimization of Mealy Machine[1L]

Minimization of incompletely specified machine–Merger Graph, Merger Table, Compatibility Graph [2L]

Lossless and Lossy Machine – Testing Table, Testing Graph [2L]

Module-3: [5L]

Regular Languages, Regular Sets, Regular Expressions, Algebraic Rules for Regular Expressions, Arden's Theorem statement and proof[1L]

Constructing Finite Automata (FA) for given regular expressions, Regular string accepted by FA[2L]

Constructing Regular Expression for a given Finite Automata[1L]

Pumping Lemma of Regular Sets. Closure properties of regular sets[1L]

Module-4: [9L]

Grammar Formalism-Context Free Grammars, Derivation trees, sentential forms. Rightmost and leftmost derivation of strings, Parse Tree, Ambiguity in context free grammars. [1L]

Minimization of Context Free Grammars. [1L], Removal of null and unit production[1L] Chomsky normal form and Greibach normal form. [1L]

Pumping Lemma for Context Free Languages. [1L]

Enumeration of properties of CFL, Closure property of CFL, Ogden's lemma & its applications [1L], Regular grammars—right linear and left linear grammars[1L]

Pushdown Automata: Pushdown automata, definition. Introduction to DCFL, DPDA, NCFL, NPDA[1L] Acceptance of CFL, Acceptance by final state and acceptance by empty state and its equivalence. [1L] Equivalence of CFL and PDA, inter-conversion.1L]

Module-5: [5L]

Turing Machine: Turing Machine, definition, model[1L]

Design of TM, Computable functions [1L], Church's hypothesis, counter machine [1L] Types of Turing machines [1L]

Universal Turing Machine, Halting problem[1L]

Textbook:

1. Introduction to Automata Theory Languages and Computation, Hopcroft.E. and Ullman J.D., Pearson Education.

Reference Books:

1. Formal Languages and Automata Theory, C. K. Nagpal, Oxford
2. -Switching and Finite Automata Theory, Zvi Kohavi, 2nd Edition, Tata McGraw Hill

CO-PO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								3
CO2	2	2	2	2								3
CO3	3	3	3	3								3
CO4	3	3	3	3								3
CO5	3	3	3	3					2	2		3

CO-PSO Mapping

COs	PSO1	PSO2	PSO3
CO1	3	3	3
CO2	3	3	3
CO3	3	3	3
CO4	3	3	3
CO5	3	3	3

Course Name: Object Oriented Programming using Java**Course Code: AM403****Contact: 3:1:0****Total Contact Hours: 48****Credits: 4****Prerequisite: Partial Object-Oriented Programming using C++****Course Outcomes (COs):****After attending the course students should be able to**

CO1	Design the process of interaction between Objects, classes & methods w.r.t. Object Oriented Programming.
CO2	Acquire a basic knowledge of Object Orientation with different properties as well as different features of Java.
CO3	Analyze various activities of different string handling functions with various I/O
CO4	Discuss basic code reusability feature w.r.t. Inheritance, Package and Interface.
CO5	Implement Exception handling, Multithreading and Applet (Web program in java) programming concept in Java.

Course Contents:**Module 1: [2L]Introduction:**

Object Oriented Analysis (OOA) & Object Oriented Design (OOD) - Concepts of object oriented programming language, Relationships among objects and classes-Generalization, Specialization, Aggregation, Association, Composition, links, Meta-class. [1L]; Object Oriented Programming concepts - Difference between Java and C++; Different features of Java [1L];

Module 2[10L]Java**Basics:**

Basic concepts of java programming - Advantages of java, Byte-code & JVM, Data types, Different types of Variables.[1L] ;Java Operators & Control statements [1L]; Java loops. [1L]; Array.[1L] ;Creation of class, object, method. [1L]; Constructor- Definition, Usage of Constructor, Different types of Constructor.[1L]; finalize method and garbage collection, Method & Constructor overloading. [1L]; this keyword, use of objects as parameter & methods returning objects.[1L]; Call by value & call by reference. [1L]; Static variables & methods.Nested & inner classes.[1L].

Module 3: [5L]

Basic String handling & I/O:

Basic string handling concepts- Concept of mutable and immutable string, Methods of String class- charAt(), compareTo(), equals(), equalsIgnoreCase(), indexOf(), length() , substring(). [1L]; toCharArray(),toLowerCase(), toString(), toUpperCase() , trim() , valueOf() methods, Methods of String buffer class- append(), capacity(), charAt(), delete(), deleteCharAt(). [1L];ensureCapacity(), getChars(), indexOf(), insert(), length(), setCharAt(), setLength(), substring(), toString(). [1L] ;Command line arguments, basics of I/O operations – keyboard input using BufferedReader [1L] ; Scanner class in Java I/O operation [1L];

Module 4: [8L]**Inheritance and Java Packages:**

Inheritance - Definition, Advantages, Different types of inheritance and their implementation. [1L]
 ;Super and final keywords, super() method. [1L]; Method overriding, Dynamic method
 dispatch.[1L]; Abstract classes & methods.[1L]; Interface - Definition, Use of Interface.[1L];
 Multiple inheritance by using Interface.[1L] ;Java Packages -Definition, Creation of packages. [1L];
 Java Access Modifiers - public, private, default and protected, Importing packages, member access
 for packages. [1L]

Module 5: [11L]**Exception handling, Multithreading and Applet Programming :**

Exception handling - Basics, different types of exception classes.Difference between Checked &
 Unchecked Exception.[1L]; Try & catch related case studies.[1L]; Throw, throws & finally. [1L];
 Creation of user defined exception. [1L]; Multithreading - Basics, main thread [1L]; Thread life
 cycle.[1L]; Creation of multiple threads-yield(), suspend(), sleep(n), resume(), wait(), notify(), join(),
 isAlive().[1L] ;Thread priorities, thread synchronization.[1L];Interthread communication, deadlocks
 for threads[1L]; Applet Programming - Basics, applet life cycle, difference between application &
 applet programming[1L]; Parameter passing in applets. [1L]

Textbooks:

1. Herbert Schildt – "Java: The Complete Reference " – 9th Ed. – TMH
2. E. Balagurusamy – " Programming With Java: A Primer " – 3rd Ed. – TMH.

Reference Books:

1. R.K Das – " Core Java for Beginners " – VIKAS PUBLISHING.
2. Rambaugh, James Michael, Blaha – " Object Oriented Modelling and Design " – Prentice Hall,
 India.

CO-PO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	2							
CO2	3	3	3	3	2							
CO3	3	3	3	3	2							
CO4	3	3	3	3	2							
CO5	3	3	3	3	2							

CO-PSO Mapping

COs	PSO1	PSO2	PSO3
CO1	3	3	3
CO2	3	3	3
CO3	3	3	3
CO4	3	3	3
CO5	3	3	3

Course Name: Probability and Statistics

Course Code: M(AM)401

Contacts: 3:0:0

Total Contact Hours: 36

Credits: 3

Prerequisite:

The students to whom this course will be offered must have the concept of (10+2) standard algebra and calculus.

Course Outcome(s):

After completion of the course students will be able to

CO1: Recall the distinctive principles of probability and statistics.

CO2: Understand the theoretical workings of theory of probability and tests of hypotheses.

CO3: Apply statistical methods to compute and explain point estimators and interval estimators for mean, variance and proportion.

CO4: Analyze statistical data from engineering experiments.

Course Content

Module 1 (Probability and Random Variables) (15 Hours)

The axioms of probability, Conditional probability, Baye's theorem, Discrete and continuous random variables, probability mass, probability density and cumulative distribution functions, mathematical expectation, Moments, Moment generating functions, Binomial, Poisson, Geometric, Uniform, Exponential and Normal distributions.

Module 2 (Two dimensional random variables) (5 Hours)

Joint distributions, Marginal and conditional distributions, Covariance, Correlation and linear regression, Transformation of random variables, Central limit theorem (for independent and identically distributed random variables).

Module 3 (Sampling Distribution) (3 Hours)

Distributions of the sample mean and the sample variance for a normal population, Chi-Square, t and F distributions, problems

Module 4 (Estimation) (4 Hours)

Unbiasedness, consistency, the method of moments and the method of maximum likelihood estimation, confidence intervals for parameters in one sample and two sample problems of normal populations, confidence intervals for proportions, problems.

Module 5 (Testing of Hypotheses) (9 Hours)

Null and alternative hypotheses, the critical and acceptance regions, two types of error, power of the test, the most powerful test and Neyman-Pearson Fundamental Lemma, tests for one sample and two sample problems for normal populations, tests for proportions, Chi square goodness of fit test and its applications, problems.

Project Domains:

1. Construction of Univariate and Bivariate frequency tables
2. Diagrammatic and Graphical representation of data.
3. Fitting of discrete and Continuous distributions

4. Regression Analysis
5. Curve Fitting
6. Tests of significance with regard to Single Mean, Two Means
7. Construction of Confidence intervals for Mean, Variance and Proportion

Text and Reference Books:

1. Sheldon M. Ross, –Introduction to Probability and Statistics for Engineers and Scientists, Academic Press, (2009).
2. D. C. Montgomery and G.C. Runger, –Applied Statistics and Probability for Engineers, 5th edition, John Wiley & Sons, (2009)..
3. Robert H. Shumway and David S. Stoffer, –Time Series Analysis and Its Applications with R Examples, Third edition, Springer Texts in Statistics, (2006).
4. N. G. Das: Statistical Methods, TMH.
5. Sancheti , D. S. & Kapoor , V.K. : *Statistics Theory , Method & Application*, Sultan chand & sons , NewDelhi
6. N.K.Dutta (2004). Fundamentals of Biostatistics, Kanishka Publishers.

CO-PO Mapping:

Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1	-	-	-	-	-	-	-	-	1
CO2	3	2	1	-	-	-	-	-	-	-	-	2
CO3	3	2	2	-	-	-	-	-	-	-	-	1
CO4	3	3	2	3	-	-	-	-	-	-	-	2

Course Name: Principles of Management
Course Code: HU(AM)401
Contacts: 2:0:0
Total Contact Hours: 24
Credits: 2

Course Outcomes (COs):

After attending the course students should be able to

CO1	Understand the basic concepts and technologies used in the field of management information System.
CO2	Have the knowledge of the different types of management information systems
CO3	Understand the processes of developing and implementing information systems.
CO4	Be aware of the ethical, social, and security issues of information systems.
CO5	An ability to effectively integrate IT-based solutions into the user environment

Course Content:

Module-1: Management Concepts: Definition, roles, functions and importance of Management, Evolution of Management thought-contribution made by Taylor,Fayol,Gilbreth,Elton Mayo,McGregor,Maslow(4L)

Module - 2: Planning and Control: Planning: Nature and importance of planning, -types of planning, Levelsof planning - The Planning Process. –MBO, SWOT analysis, McKinsey’s7S Approach. Organizing for decision making: Nature of organizing, span of control, Organizational structure – line and staffauthority. Basic control process -control as a feedback system – Feed Forward Control –Requirements for effectivecontrol – control (4L)

Module - 3: Group dynamics: Types of groups, characteristics, objectives of Group Dynamics. Leadership: Definition, styles & functions of leadership, qualities for good leadership, Theories of leadership(4L)

Module – 4: Work Study and work measurement: Definition of work study, Method Study Steps, Tools and Techniques used in the Method Study and Work Measurement Time Study: Aim & Objectives, Use of stopwatch procedure in making Time Study. Performance rating, allowances and its types. Calculation of Standard Time. Work sampling(4L)

Module - 5: Marketing Management: Functions of Marketing, Product Planning and development, Promotional Strategy (2L)

Module - 6: Quality management: Quality definition, Statistical quality control, acceptance sampling ,Control Charts –Mean chart, range chart,c chart,p chart,np chart, Zero Defects, Quality circles,, Kaizen & Six Sigma,ISO -9000 Implementation steps, Total quality management (6L)

Text Books:

1. Essentials of Management, by Harold Koontz & Heinz Weihrich Tata McGraw
2. Production and Operations Management-K.Asathapa,K .Shridhara Bhat,Himalayan Publishing House

References:

1. Organizational Behavior, by Stephen Robbins Pearson Education, New Delhi
2. New era Management, Daft, 11th Edition, Cengage Learning
3. Principles of Marketing, Kotlar Philip and Armstrong Gary, Pearson publication

CO-PO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2							3	
CO2	3	2	2	2							3	
CO3	3	3	2	2							3	
CO4	3	2	2	2							3	
CO5	3	2	2	2							3	

CO-PSO Mapping

COs	PSO1	PSO2	PSO3
CO1	2	2	2
CO2	2	2	2
CO3	2	2	2
CO4	2	2	2
CO5	2	2	2